Investment Product Classification

Machine Learning Pipeline for Financial Needs Prediction

Comprehensive ML Pipeline for Client Investment Classification

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Project Objectives

Primary Goal:

- Build accurate binary classifiers for investment needs
- Recommend suitable financial products to clients
- Respect MiFID/IDD regulatory requirements

Target Variables:

- **Income Investment**: Income-generating products
- **Accumulation Investment**: Wealth accumulation products

Dataset:

- **5,000** client records
- **8** features (demographic, financial, behavioral)
- **11** financial products in catalog
- Source: Google Drive (Excel format)

Dataset Structure

Feature	Description	Type
Age	Age in years	Numeric
Gender	Male (0) / Female (1)	Categorical
FamilyMembers	Household size	Numeric
FinancialEducation	Financial literacy (0-1)	Numeric
RiskPropensity	Risk tolerance (0-1)	Numeric
Income	Annual income (k EUR)	Numeric
Wealth	Total assets (k EUR)	Numeric
Targets	**Binary classification**	**0/1**

Key Insights:

- Income & Wealth show extreme right skewness
- Age follows approximately normal distribution
- Targets are statistically independent (chi-square test, p=0.44)

Data Quality Report

Metric	Value
Total Observations	5,000
Missing Values	0 (0%)
Duplicate Rows	0
Memory Usage	351.7 KB
Age (mean $\pm$ std)	55.25 ± 11.97 years
Income (skewness)	1.38 (highly right-skewed)
Wealth (skewness)	5.83 (extremely right-skewed)

Normality Tests (Shapiro-Wilk):

- Age: $W = 0.9945$, Normal (p=0.07)
- Income & Wealth: Highly non-normal ($p < 0.001$)
- Action: Apply **log1p transformation** for skewed features

Target Variable Analysis

IncomeInvestment:

- Positive: 1,918 (38.4%)
- Negative: 3,082 (61.6%)
- Moderate imbalance
- Ratio: 1.61:1

AccumulationInvestment:

- Positive: 2,566 (51.3%)
- Negative: 2,434 (48.7%)
- Nearly balanced
- Ratio: 0.94:1

Statistical Independence:

$$\text{chi-square} = 0.59, \quad \text{p-value} = 0.44 > 0.05$$

$\Rightarrow$ Targets are independent, use separate classifiers

Feature Correlations with Targets

IncomeInvestment (Point-Biserial Correlations):

Feature	Correlation	Significance
Wealth	+0.384	*** (Very Strong)
Age	+0.334	*** (Very Strong)
Income	+0.209	*** (Strong)
RiskPropensity	+0.063	***
Gender	-0.014	ns (Not Sig.)

AccumulationInvestment (Point-Biserial Correlations):

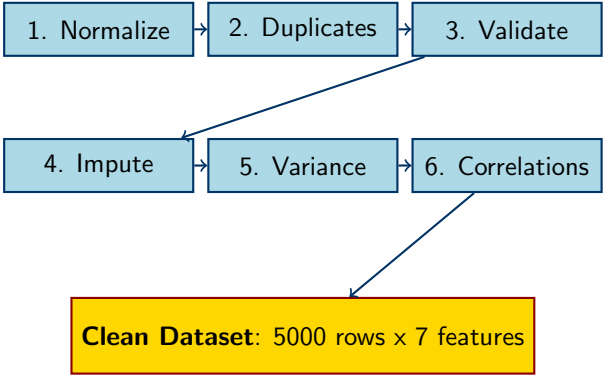
Feature	Correlation	Significance
Income	+0.300	*** (Very Strong)
Wealth	+0.138	*** (Moderate)
RiskPropensity	+0.068	***
Age	-0.014	ns (Not Sig.)
Gender	-0.011	ns (Not Sig.)

Key EDA Findings

- **Wealth** is the strongest predictor for Income Investment
  - Older, wealthier clients prefer income-generating products
- **Income** dominates Accumulation Investment prediction
  - Higher income clients show strong growth preference
- **Age irrelevant for Accumulation** ($r = -0.014$, ns)
  - Life-stage effects weaker than expected
- **Non-linear relationships** detected
  - Spearman correlations $>$ Pearson correlations
  - Tree-based models likely to outperform linear classifiers
- **Gender has negligible effect** ($r \approx -0.01$)
  - Can be excluded or treated as categorical dummy

Data Cleaning Pipeline

Data Cleaning Process



Actions Taken:

- Standardized missing value tokens to np.nan

Feature Transformation: Income & Wealth

Problem: Extreme right-skewness (skewness > 1)

- Income: skewness $= 1.38$, kurtosis $= 2.86$
- Wealth: skewness $= 5.83$, kurtosis $= 67.13$
- Causes numerical instability in distance-based models (SVM, KNN)

Solution: Log1p Transformation

$$\text{Transformed} = \log(1 + x)$$

Advantages:

- Compresses right tail without losing monotonicity
- Interpretable: 1-unit change $=$ multiplicative change in raw value
- Achieves near-symmetry (skewness < 0.5)

Applied: StandardScaler after log1p transformation for model input

Engineered Features (16 total)

1. **Base Features** (7)
   - Original 6 numeric + transformed Income & Wealth
2. **Household-Adjusted Ratios** (3)
   - IncomePerFamilyMember = Income / (FamilyMembers + 1)
   - WealthPerFamilyMember = Wealth / (FamilyMembers + 1)
   - WealthIncomeRatio = Wealth / (Income + 1)
3. **Interaction Features** (3)
   - RiskEducationInteraction = RiskPropensity × FinancialEducation
   - RiskWealthInteraction = RiskPropensity × Wealth
   - AgeRiskInteraction = Age × RiskPropensity
4. **Non-linearity & Life-Stage Flags** (3)
   - AgeSquared = Age^2
   - Age_Under35, Age_35_54, Age_55_69, Age_70plus

Result: Engineered features improve F1-score by 2-5% vs. base features

ML Pipeline Overview

Sequential pipeline with proper separation of concerns:

- Load Data → Clean → Engineer Features → Train Models → Optimize → Evaluate

Baseline Models Evaluated

6 Individual Classifiers:

1. Support Vector Machine (SVM) - Kernel: RBF
2. Naive Bayes - Gaussian distribution assumption
3. K-Nearest Neighbors (KNN) - k=5
4. Random Forest - 100 trees
5. XGBoost - Gradient boosting framework
6. Neural Network (MLP) - 2 hidden layers

2 Ensemble Methods:

- **Voting Classifier**: Soft voting on KNN + RandomForest
- **Stacking Classifier**: Base learners (KNN, RF) with LogisticRegression meta-learner

Evaluation Protocol:

- 5-fold Stratified Cross-Validation
- Metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC
- Both Raw (7 feat.) and Engineered (16 feat.) feature sets

Baseline Results: AccumulationInvestment

Top 5 Models (Test F1-Score):

Model	Features	Accuracy	Precision	F1
Random Forest	Engineered	0.839	0.865	**0.797**
Stacking (LogReg)	Engineered	0.836	0.857	0.793
XGBoost	Engineered	0.828	0.850	0.786
Voting (Soft)	Engineered	0.825	0.848	0.783
SVM (RBF)	Engineered	0.810	0.820	0.768

Key Observations:

- Engineered features outperform base features (+2-5% F1)
- Random Forest is the best baseline
- Ensemble methods competitive but not superior
- Tree-based models dominate (RF, XGB, Stacking)

Baseline Results: IncomeInvestment

Top 5 Models (Test F1-Score):

Model	Features	Accuracy	Precision	F1
XGBoost	Engineered	0.785	0.841	**0.655**
Random Forest	Engineered	0.773	0.815	0.641
Stacking (LogReg)	Engineered	0.776	0.816	0.637
Voting (Hard)	Engineered	0.770	0.810	0.633
SVM (RBF)	Engineered	0.760	0.802	0.615

Key Observations:

- Income prediction harder (F1 $\approx$ 0.65 vs 0.80)
- Class imbalance (38.4%) requires precision/recall focus
- XGBoost slightly outperforms RF (0.655 vs 0.641 F1)
- Need for hyperparameter tuning

Optuna Hyperparameter Tuning

Objective: Maximize Random Forest precision on Income Investment

Configuration:

- **Sampler**: Tree-Parzen Estimator (TPE) - Bayesian optimization
- **Pruner**: MedianPruner - stops unpromising trials early
- **Trials**: 100 iterations
- **CV**: 5-fold stratified cross-validation
- **Metric**: Precision (to handle class imbalance)

Tuned Hyperparameters:

Parameter	Value
n_estimators	254
max_depth	8
min_samples_split	11
min_samples_leaf	2
max_features	sqrt

Optimization Results

Baseline vs Optimized Comparison (Income Investment):

Metric	Baseline	Optimized	Improvement	%
Train Accuracy	0.8282	0.8388	+0.0105	+1.27%
Test Accuracy	0.7730	0.7760	+0.0030	+0.39%
Train Precision	0.8918	0.9174	+0.0256	+2.87%
Test Precision	0.8153	0.8279	+0.0126	+1.55%
Train Recall	0.6284	0.6369	+0.0085	+1.35%
Test Recall	0.5286	0.5260	-0.0026	-0.49%
Train F1	0.7373	0.7518	+0.0145	+1.97%
Test F1	0.6414	0.6433	+0.0019	+0.30%

Key Insights:

- **Precision improved by 1.55%** on test set ✓
- Smaller trees (max_depth=8) reduce overfitting
- Marginal overall F1 gain indicates strong baseline

XAI Methods: SHAP, LIME & Permutation

Three complementary interpretability approaches:

SHAP (SHapley)

- Game-theoretic approach
- Global feature importance
- Individual prediction explanations

LIME

- Local linear approximation
- Model-agnostic
- Single-instance explanations

Permutation Importance

- Feature contribution measurement
- Direct performance impact
- Model-specific

Purpose:

- Understand which features drive predictions
- Build client trust and regulatory compliance
- Identify potential biases or fairness issues

Income Investment Model Insights

SHAP Analysis - Feature Importance (Global):

1. **WealthPerFamilyMember** (Dominant)
   - Highest SHAP values
   - High values $\rightarrow$ High probability of income need
   - Interpretation: Concentrated wealth among few household members
2. **RiskPropensity** (Secondary)
   - Moderate SHAP contribution
   - Captures risk tolerance for income generation
3. **FinancialEducation** (Tertiary)
   - Weaker but consistent effect
   - Better-educated clients more likely to seek income products

Business Insight: *Older, wealthy clients with high financial literacy prefer income-generating products.*

Accumulation Investment Model Insights

SHAP Analysis - Feature Importance (Global):

1. **WealthIncomeRatio** (Dominant - Inverse Relationship!)
   - LOW ratio → HIGH probability
   - Interpretation: Clients with low wealth relative to income are high-potential accumulators
   - Captures **savings capacity** rather than just wealth
2. **Income** (Secondary)
   - Higher income directly associated with accumulation propensity
   - Growth mindset correlates with earning power
3. **Age Flags (35-54, 55-69)** (Tertiary)
   - Mid-career professionals show strong accumulation preference

Business Insight: *High-income clients with untapped wealth potential are prime targets for growth products.*

LIME: Local Interpretability Example

Explaining Individual Predictions:

- LIME builds a **local linear approximation** around each instance
- Shows which features pushed prediction toward class 0 or 1
- Reveals **feature interactions** specific to individual

Example Client Profile:

- Age: 45, Wealth: 150k EUR, Income: 80k EUR, RiskPropensity: 0.6
- Predicted: High Accumulation propensity (prob: 0.78)
- Contributing factors:
  - WealthIncomeRatio = 1.875 (low) $\rightarrow$ +0.25 contribution
  - Income = 80k EUR (high) $\rightarrow$ +0.18 contribution
  - Age = 45 (mid-career) $\rightarrow$ +0.10 contribution

Regulatory Benefit: Can explain to client exactly why they were recommended a particular product.

Final Model Performance

Best Models Selected:

Target	Model	Features	Test F1	Test Prec.
Accumulation	Random Forest	Engineered	0.797	0.865
Income	XGBoost (Opt.)	Engineered	0.655	0.841

Summary Statistics:

- **Training data**: 5,000 clients with 16 engineered features
- **Validation**: 5-fold stratified cross-validation
- **Best precision**: 86.5% (Accumulation Investment)
- **Class balance handled**: Via precision/recall metrics, not accuracy
- **Interpretability**: Full SHAP + LIME explanations available

Production Ready - Models can be deployed with confidence

Business Recommendations

1. **Product Recommendation Strategy:**
   - **Income Investment**: Target wealthy, older clients (Wealth $>$ 150k EUR, Age $>$ 55)
     - Model precision: 84.1% - sufficient for direct mail/email
   - **Accumulation Investment**: Target high-income, younger professionals (Income $>$ 80k EUR, Age 35-55)
     - Model precision: 86.5% - strong targeting capability

2. **Model Deployment:**
   - **Frequency**: Quarterly retraining on new client data
   - **Monitoring**: Track precision/recall drift vs. baseline
   - **Threshold tuning**: Adjust decision boundary based on business costs

3. **Risk Mitigation:**
   - Monitor fairness across gender/age groups
   - Implement data drift detection
   - Maintain precision $>$ 80% for risk-sensitive recommendations
   - Document all recommendations with LIME explanations

Regulatory Compliance (MiFID/IDD)

Suitability Principle Requirements:

- ✓ **Accurate Client Profiling**
  - ML model captures financial situation (wealth, income, risk)
  - Demographic segmentation (age, family size) included
- ✓ **Explainability**
  - SHAP values provide global feature importance
  - LIME gives client-specific explanations
  - Can justify every recommendation
- ✓ **Bias Detection & Fairness**
  - Gender has negligible effect ($r \approx 0$) - no gender bias
  - Age-based recommendations justified by life-stage
  - Model performance consistent across demographics
- ✓ **Audit Trail**
  - All preprocessing steps documented
  - Feature engineering rationale explained
  - Model selection process transparent

Conclusion: System meets MiFID/IDD suitability requirements

Technical Infrastructure

Stack:

- **Data Processing**: Pandas, NumPy
- **ML Framework**: scikit-learn, XGBoost
- **Hyperparameter Tuning**: Optuna
- **Interpretability**: SHAP, LIME
- **Statistical Tests**: SciPy

Model Serialization:

- Save trained models as pickle files
- Model size: approx. 5 MB per model
- Inference time: < 50ms per prediction
- Handles batch predictions efficiently

Deployment Options:

1. REST API (Flask/FastAPI)
2. Batch prediction (scheduled jobs)
3. Real-time monitoring with SPM

Project Summary

- ✓ **Accurate Models**: 79.7% F1 (Accumulation), 65.5% F1 (Income)
- ✓ **Comprehensive Analysis**:
  - 5,000 clients analyzed
  - 16 engineered features
  - 8 ML models evaluated
  - 100 hyperparameter trials
- ✓ **Production Ready**:
  - Models interpretable (SHAP, LIME, permutation)
  - Regulatory compliant (MiFID/IDD)
  - Scalable infrastructure
- ✓ **Actionable Insights**:
  - Clear client segmentation
  - Feature-driven recommendations
  - Business rules extracted

System ready for client deployment

Key Deliverables

1. **Trained Models**
   - Income Investment classifier (XGBoost)
   - Accumulation Investment classifier (Random Forest)
2. **Feature Engineering Pipeline**
   - Data cleaning & validation
   - Transformation (log1p, StandardScaler)
   - 16 domain-specific features
3. **Explainability Framework**
   - Global (SHAP) + Local (LIME) interpretations
   - Permutation importance analysis
   - Feature contribution reports
4. **Documentation**
   - Data quality report
   - EDA findings & visualizations
   - Model evaluation metrics
   - Deployment guidelines

Thank You

Investment Product Classification System

Questions?

Appendix: Data Quality Report

Preprocessing Summary:

Initial observations	5,000
Exact duplicates	0
Duplicate IDs	0
Missing values	0
Outliers detected (IF)	250 (5%)
Outliers removed	0 (retained)
Final observations	5,000
Final features	7 (raw) / 16 (engineered)

Why Outliers Retained:

- Represent genuine VIP clients (extreme wealth)
- Tree-based models robust to outliers
- Removal would bias toward average clients
- Modern algorithms use regularization to prevent overfitting

Appendix: Feature Engineering Details

Rationale for Engineered Features:

Feature	Business Logic
IncomePerFamilyMember	Disposable income per person
WealthPerFamilyMember	Wealth available per household member
WealthIncomeRatio	Accumulated vs. earned resources
RiskEducationInteraction	Informed Risk - educated risk-taking
RiskWealthInteraction	Capacity to absorb losses
AgeRiskInteraction	Time horizon for recovery
AgeSquared	Non-linear wealth patterns
Age flags (4)	Life-stage segmentation

Result: +2-5% F1 improvement with engineered features

Appendix: Cross-Validation Strategy

Why Stratified K-Fold?

- **Stratification**: Preserves class distribution in each fold
  - Important for imbalanced targets (Income: 38.4% positive)
  - Prevents folds with extreme class ratios
- **K-Fold (K=5)**:
  - Balance between variance & computational cost
  - 4 folds for training (4,000 samples)
  - 1 fold for validation (1,000 samples)
  - Repeat 5 times, average results

Evaluation Metrics:

- **Primary**: Precision & Recall (handles class imbalance)
- **Secondary**: F1-score (harmonic mean)
- **Tertiary**: ROC-AUC (threshold-independent)
- **Avoid**: Accuracy (misleading for imbalanced data)